

Large cardinals at the inconsistency frontier

A Prikry consistency refinement and
local definability obstructions

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Principal result developed here. Let $j : V_\Lambda \rightarrow V_\Lambda$ be an $\text{I3}_{\text{wf}(0)}$ -embedding, let $\kappa = \text{crit}(j)$, and let U be its derived normal ultrafilter. Prikry forcing with U makes κ κ -cover exacting. Consequently,

$$\text{Con}(\text{ZFC} + \text{I3}_{\text{wf}(0)}) \implies \text{Con}(\text{ZFC} + \exists \lambda \text{ CEx}_\lambda(\lambda)).$$

This replaces the supplied report's explicit I2 upper bound by a weaker rank-into-rank assumption. The construction combines the well-founded-part localization used for ordinary exactingness with a finite-change family of Prikry-name evaluations. A separate tree argument puts the required local embedding *inside* the constructed model.

Positive and negative conclusions

Relative consistency. Sections 3–5 give the construction and its internalization argument, including the uniform cardinal bound on the covering family.

Conditional inconsistency. If λ is exacting, no nonempty OD_{V_λ} family of fewer than λ short cofinal maps into λ exists. In fact, one coordinate projection of any such definable family has order type λ . Individual maps need not be definable.

Scope. These results do not settle cover exactingness above a strongly compact cardinal, ordinary exactingness above an extendible, or the unrestricted cardinal-preserving embedding problem.

Provenance and status. This is a research draft with detailed proofs, not a claim of an independently refereed discovery. The iteration facts are from Andretta–Dimonte and Aguilera–Bagaria–Goldberg–Lücke. Blue–Goldberg's 2026 notes already anticipate cover versions of related constructions. The precise refinement below is established here as a continuation of the supplied report; its priority in the literature is not certified. No proof-assistant verification is claimed.

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Reading route

The main forcing theorem is Theorem 5.2; the two reusable arguments are the internalization lemma, Lemma 3.1, and the finite-change cloud lemma, Lemma 4.1. The local negative result is Theorem 7.1. Section 9 records the proof's sensitive interfaces and the questions it leaves open.

1 What this continuation adds

The supplied unified report separates an axiom from an additional closure, definability, or stationarity requirement that might make it contradictory. It also singles out cover exactingness as a useful boundary: its existence has a consistency construction, while placing it above an extendible cardinal is ruled out in the cited lecture notes. We keep that organization, but pursue a positive construction and a more flexible local obstruction rather than repeat the candidate survey.[1, 2]

The starting comparison in the supplied report was $\text{Con}(\text{ZFC} + \text{I2}) \Rightarrow \text{Con}(\text{ZFC} + \exists \lambda \text{CE}_{\lambda}(\lambda))$. The ordinary-exacting construction in Aguilera–Bagaria–Goldberg–Lücke uses only $\text{I3}_{\text{wf}(0)}$. [3, Theorem 5.4] The mathematical task is to retain a *uniformly small predicate* containing an arbitrary object, not just an unexpanded exacting witness.

Main theorem. Suppose $j : V_{\Lambda} \rightarrow V_{\Lambda}$ is an $\text{I3}_{\text{wf}(0)}$ -embedding with critical point κ , and

$$U = \{A \subseteq \kappa : \kappa \in j(A)\}.$$

If G is generic for Prikry forcing $\mathbb{P}_U$, then

$$V[G] \models \text{CE}_{\kappa}(\kappa).$$

In particular, the existence of an $\text{I3}_{\text{wf}(0)}$ -embedding implies the existence of a countable transitive model of ZFC with a λ -cover-exacting cardinal.

The last assertion is stronger than an informal appeal to a generic over the universe. Its proof passes to a countable elementary submodel after establishing the forcing theorem; see Corollary 6.1. No converse consistency reduction is asserted.

The second result strengthens the original report’s single-definable-map obstruction. Write $\text{OD}_{V_{\Lambda}}$ for definability from ordinal parameters and finitely many parameters belonging to V_{Λ} .

Definable-family theorem. Let λ be exacting and let $0 < \rho < \lambda$. Suppose $F \in \text{OD}_{V_{\Lambda}}$ is a nonempty family of cofinal maps $\rho \rightarrow \lambda$. Then for some $\xi < \rho$,

$$\text{ot}\{f(\xi) : f \in F\} = \lambda.$$

Consequently $|F| \geq \lambda$ in the ambient universe. If every member of F is increasing, all coordinate projections from some coordinate onward have order type λ .

The hypothesis is that the *family* is definable. It is not assumed that a member of the family belongs to HOD, that the family has a definable well-order, or that a definable choice function selects one of its members. This distinction is what makes the obstruction stronger than simply asking for a cofinal map in $\text{HOD}_{V_{\Lambda}}$.

1.1 A precise novelty ledger

Ingredient or conclusion	Status in this continuation
Iteration and well-founded-part facts	Imported mathematical inputs, with the exact source locators recorded below.
The I2 cover construction	Already in Blue–Goldberg’s notes; not presented as a discovery here.
The $I3_{\text{wf}(0)}$ cover refinement	An explicit proof developed here by combining and extending those arguments. The notes’ broader anticipation of cover variants prevents a justified priority claim.
The definable-family obstruction	A proved consequence of the established HOD-regularity theorem, with a quantitative projection bound. It is not a new refutation of bare exactingness.
The smaller-strongly-compact case	Unresolved by these arguments. No preservation of such a cardinal below the new exacting cardinal is proved.

A distinction between a new deduction in this document and a new theorem in the historical literature is essential. The latter requires a more exhaustive priority investigation and specialist scrutiny than a targeted source check can provide. The mathematical claims below are stated with their hypotheses and proofs, rather than with a claim of priority.

2 Definitions and the iteration interface

2.1 Cover exactingness and its quantifiers

Definition 2.1. For $\alpha > \lambda$ and $Y \subseteq V_\alpha$, let $\text{RelEx}(\lambda, \alpha, Y)$ assert that there are X and an elementary map

$$k : (X, \in, Y \cap X) \longrightarrow (V_\alpha, \in, Y)$$

with

$$(X, \in, Y \cap X) < (V_\alpha, \in, Y), \quad V_\lambda \cup \{\lambda\} \subseteq X, \quad k(\lambda) = \lambda, \quad k \upharpoonright \lambda \neq \text{id}.$$

A cardinal λ is γ -cover exacting if

$$\forall \alpha > \lambda \forall y \in V_\alpha \exists Y \subseteq V_\alpha \left[y \in Y \wedge |Y| \leq \gamma \wedge \text{RelEx}(\lambda, \alpha, Y) \right]. \quad (1)$$

This is the bounded-predicate definition used in the supplied report and in Blue–Goldberg’s notes.[2, Definitions 2.6 and 3.1] The object y is a *member* of Y . It need not belong to X , and the witness need not fix y . The bound γ is independent of α and y . The substructure X need not be transitive. Every occurrence of this definition in a model W uses V_α^W , cardinality in W , and witnesses belonging to W .

Ordinary exactingness, $\text{Ex}(\lambda)$, requires the same witness after omitting the predicate Y , for every cardinal height $\alpha > \lambda$. Every cover-exacting cardinal is therefore exacting.

The language is finite, so satisfaction for these set structures is available in ordinary set theory. Thus (1) is a first-order set-theoretic assertion, not a proper-class embedding axiom.

2.2 The weak iterability hypothesis

Fix $j : V_\Lambda \rightarrow V_\Lambda$ with critical sequence

$$\kappa_0 = \text{crit}(j), \quad \kappa_{n+1} = j(\kappa_n), \quad \Lambda = \sup_{n < \omega} \kappa_n.$$

All uses of $I3$ in the construction have this normalization. The canonical extension on a subset $A \subseteq V_\Lambda$ is

$$j^+(A) = \bigcup_{\beta < \Lambda} j(A \cap V_\beta).$$

Let $j_{0,1} = j$, let $j_{n+1,n+2} = j^+(j_{n,n+1})$, and form the commuting finite iteration system $\langle j_{m,n} : m \leq n < \omega \rangle$. Its direct limit is denoted M_ω^j , with maps $j_{n,\omega}$. The limit may be ill-founded. Its well-founded part, identified with its transitive collapse, is W_ω^j .

Definition 2.2. The embedding j is an $I3_{\text{wf}(0)}$ -embedding if

$$j_{0,\omega}[\kappa_0^+] \subseteq W_\omega^j. \quad (2)$$

This is Definition 5.2 of Aguilera–Bagaria–Goldberg–Lücke with $n = 0$. [3] It does *not* require the whole direct limit to be well-founded. In particular, it is not the same hypothesis as $I3_1$, which requires full well-foundedness of the first ω -iterate.

2.3 The imported interface, rather than an assumed extension

We use the following iteration facts. The finite-iteration construction and amenable-predicate elementarity are treated by Andretta–Dimonte; the fixed-code and well-founded-part formulations below are recorded in Aguilera–Bagaria–Goldberg–Lücke. [4, Section 3, especially Lemma 3.3] [3, Proposition 5.1 and proof of Theorem 5.4]

Proposition 2.3 (Iteration interface). With the preceding notation:

1. $V_\Lambda \cup \{\Lambda\} \subseteq W_\omega^j$, $j_{n,\omega} \upharpoonright V_{\kappa_n} = \text{id}$, and $j_{n,\omega}(\kappa_n) = \Lambda$.
2. If $A \in V_{\kappa_0+1}$, then $j_{0,\omega}(A) \in V_{\Lambda+1} \cap W_\omega^j$ and $j^+(j_{0,\omega}(A)) = j_{0,\omega}(A)$.
3. If $E \subseteq \Lambda \times \Lambda$ and $j^+(E) = E$, then $j \upharpoonright \Lambda : (\Lambda, E) \rightarrow (\Lambda, E)$ is elementary.
4. Put $U_0 = \{A \subseteq \kappa_0 : \kappa_0 \in j(A)\}$ and $U = j_{0,\omega}(U_0)$. For $A \in \mathcal{P}(\Lambda) \cap W_\omega^j$,

$$A \in U \iff \exists m \forall n \geq m (\kappa_n \in A). \quad (3)$$

Clause (3) concerns the *expanded* structure, not an assertion that j^+ is an $I1$ -embedding. One applies amenable-predicate elementarity to (V_Λ, E) and then interprets the ordinal domain. Every bounded piece of E belongs to V_Λ , and $\text{cf}(\Lambda) = \omega$ supplies the relevant hypothesis of the cited lemma.

For clause (4), a subset represented at a finite stage has its eventual membership along the critical sequence determined by that stage's derived normal measure. If it is measure one, all sufficiently late critical points belong to it; if it is measure zero, the complementary set is measure one. This also explains why (3) is an *external* description of an internal ultrafilter. It does not make Λ measurable in the ambient universe, where $\text{cf}(\Lambda) = \omega$.

These are the only rank-into-rank iteration inputs needed below. The new internalization argument does not assume an extension of j to an arbitrary higher rank or to the universe.

3 Internalizing an external local witness

The main obstacle is not the existence of an external map. A forcing construction has to produce a model that *contains* the embedding witnesses in Definition 2.1. The following lemma handles this transfer, including high-rank predicates.

Lemma 3.1 (Enumerated-domain internalization). Work in an ambient universe satisfying ZFC. Let W be a transitive set model of a sufficiently strong finite fragment of ZFC. Suppose there are an external elementary map $e : W \rightarrow W$, an infinite cardinal λ of W , and $c = \langle v_n : n < \omega \rangle \in W$ such that

1. c is strictly increasing and cofinal in λ , $e(\lambda) = \lambda$, and $e(v_n) = v_{n+1}$;
2. $\omega < v_0$, $V_\lambda^{\text{ambient}} \subseteq W$, and $W \models |V_\lambda| = \lambda$;
3. $\alpha > \lambda$ and $Y \subseteq V_\alpha^W$ belong to W , with $e(\alpha) = \alpha$ and $e(Y) = Y$.

Then

$$W \models \text{RelEx}(\lambda, \alpha, Y).$$

It is not assumed that $e \in W$ or that $e \upharpoonright V_\lambda \in W$.

Proof. All the structures, functions, and trees that are said to be chosen in W below are sets of W . Write

$$\mathcal{A} = (V_\alpha^W, \in, Y).$$

Inside W , choose an elementary substructure $X < \mathcal{A}$ of cardinality λ containing $V_\lambda^W \cup \{\lambda\}$. The downward Löwenheim–Skolem construction applies to the finite language with its unary predicate. Set $Z = e(X)$, with the induced expanded structure. Since $e(\mathcal{A}) = \mathcal{A}$,

$$W \models Z < \mathcal{A}.$$

In W , take a bijection $f : \lambda \rightarrow X$ with $f(0) = \lambda$ and $f(1) = v_0$, and put $g = e(f) : \lambda \rightarrow Z$. Then $g(0) = \lambda$ and $g(1) = v_1$.

For $n < \omega$, define in W

$$X_n = f''v_n, \quad Z_n = g''v_{n+1}.$$

Let T be the following tree in W . A node at level n is a finite list $(p_0, \dots, p_n)$ such that, for every $m \leq n$,

1. $p_m : X_m \rightarrow Z_m$ is a partial elementary map from the full structure on X to the full structure on Z ;
2. $p_m(\lambda) = \lambda$ and $p_m(v_0) = v_1$;
3. $p_{m+1} \upharpoonright X_m = p_m$ when $m < n$.

The order is end extension of finite lists. Here “partial elementary” means agreement on every first-order formula whose finitely many parameters lie in the map’s domain. It does not mean elementarity between X_m and Z_m as separate substructures. Satisfaction for X and Z is a set in W , so this definition gives a set tree in W .

For each n , consider the external restriction

$$q_n = e \upharpoonright v_n.$$

It maps v_n into v_{n+1} , and its graph has rank strictly below λ . Hence $q_n \in V_\lambda^{\text{ambient}} \subseteq W$. Using f, g, q_n inside W , form

$$p_n(f(\xi)) = g(q_n(\xi)) \quad (\xi < v_n). \quad (4)$$

The elementary evaluation identity gives, externally,

$$g(e(\xi)) = e(f)(e(\xi)) = e(f(\xi)),$$

so $p_n = e \upharpoonright X_n$. Thus each p_n is partial elementary and has the required two prescribed values. Satisfaction for the transitive model’s set structures is absolute, so W recognizes these properties. The finite lists $(p_0, \dots, p_n)$ belong to T . They form an external infinite branch.

If W had no infinite branch through T , its well-founded recursion and dependent choice would give a rank function on the strict-extension relation of T with values in its ordinals. Since W is transitive, that would be an actual ordinal-valued rank function in the ambient universe. It would strictly decrease along the displayed external branch, which is impossible. Therefore a branch B belongs to W .

The union of the maps along B is an elementary map $k : X \rightarrow Z$ in W . Its domain is all of X , because $\sup_n v_n = \lambda$ and f is onto. Every finite tuple is contained in one X_n , which verifies full elementarity of the union. Moreover,

$$k(\lambda) = \lambda, \quad k(v_0) = v_1 \neq v_0.$$

Composing with the elementary inclusion $Z < \mathcal{A}$ proves $\text{RelEx}(\lambda, \alpha, Y)$ in W . $\square$

3.1 Why the argument is genuinely internal

Two tempting shortcuts would be wrong. The external map $e \upharpoonright X$ is not declared to belong to W . Instead, each bounded restriction is *recoded by a small ordinal map* q_n that does belong to W . Nor is the external branch simply imported into W : well-foundedness absoluteness supplies some internal branch, possibly a different one.

The finite-list definition of T is also deliberate. A map $X_n \rightarrow Z_n$ need not have its restriction to X_m land in Z_m for $m < n$. Requiring a coherent finite list, with a separate range bound at each level, makes the tree closed under initial segments and avoids this otherwise easy oversight.

The required fragment of set theory can be fixed uniformly before the application. It includes the set-satisfaction construction, the elementary hull construction, the indicated instances of Replacement and Separation, and the theorem that a well-founded set relation has an ordinal rank. Appendix A explains how this fragment is secured by reflection. No assumption that an arbitrary rank segment satisfies all of ZFC is made.

4 Finite-change clouds of Prikry names

Let N be transitive, let $D \in N$ be a normal measure on λ as computed in N , and let $\mathbb{Q} = \mathbb{P}_D^N$. For an increasing cofinal sequence $h : \omega \rightarrow \lambda$, define

$$G_h = \{(s, A) \in \mathbb{Q} : s \text{ is an initial segment of } h \text{ and } h[\omega \setminus \text{dom}(s)] \subseteq A\}.$$

The forcing order is the usual one: extend the finite increasing stem, choose its new points from the measure-one set, and shrink that set. The parameter $\mathbb{Q}$ denotes the ground partial order. It is not a newly computed forcing from all subsets of λ in the extension.

For increasing sequences c, h , write

$$h \sim c \iff |\text{ran}(h) \Delta \text{ran}(c)| < \omega. \quad (5)$$

This is equality modulo finite change of the *ranges*, not coordinatewise eventual equality of sequences. In particular, $c^- = \langle c(n+1) : n < \omega \rangle$ satisfies $c^- \sim c$.

Lemma 4.1 (Finite-change cloud). Suppose c is $\mathbb{P}_D^N$ -generic over N and $W = N[G_c]$. Let $\tau \in N$ be a $\mathbb{Q}$ -name, and suppose

$$N \models \mathbf{1}_{\mathbb{Q}} \Vdash \tau \in V_{\check{\alpha}}.$$

Define in W

$$Y_{\tau,c} = \{\tau^{G_h} : h \in {}^\omega \lambda \text{ is strictly increasing and } h \sim c\}. \quad (6)$$

Then $Y_{\tau,c} \in W$, $\tau^{G_c} \in Y_{\tau,c} \subseteq V_{\check{\alpha}}^W$, and $W \models |Y_{\tau,c}| \leq \lambda$.

If an external elementary $e : W \rightarrow W$ fixes $\lambda, \mathbb{Q}, \tau, \alpha$ and satisfies $e(c) = c^-$, then $e(Y_{\tau,c}) = Y_{\tau,c^-}$.

Proof. Every $h \sim c$ is eventually in every member of D , because c has that property. The Mathias criterion therefore makes h generic over N . This criterion and the elementary properties of Prikry forcing are used in their model-relative forms.[3, proof of Theorem 5.4]

Moreover, $N[G_h] = N[G_c]$. Indeed, $h \in W$ gives one inclusion. For the other, the two finite difference sets are finite sets of ordinals below λ , hence are elements of N . Using these sets, c is recovered from h by taking the increasing enumeration of its modified range. Thus $c \in N[G_h]$. It follows from the forcing theorem that every value in (6) belongs to the same $V_{\check{\alpha}}^W$.

The definition of $Y_{\tau,c}$ inside W uses only $\tau, \mathbb{Q}, c$ and λ . It does *not* quantify over N or require that N be an element of W . A forcing-name evaluation is a well-founded recursive operation on the name and the set G_h . Replacement in W therefore forms the displayed family.

An $h \sim c$ is determined by two finite subsets of λ : points deleted from $\text{ran}(c)$ and points added to it. There are at most λ such pairs in W , since λ is an infinite cardinal. Consequently there are at most λ possible h and at most λ evaluations. The choice $h = c$ includes the required object.

Finally, elementarity applied to the internal definition gives

$$e(Y_{\tau,c}) = Y_{e(\tau),e(c)} = Y_{\tau,c^-}.$$

The relation $\sim$ is an equivalence relation, and $c^- \sim c$, so the defining families of sequences are identical. Hence $Y_{\tau,c^-} = Y_{\tau,c}$. $\square$

Remark 4.2. The value $y = \tau^{G_c}$ need not be fixed. In fact, $e(y) = \tau^{G_{c^-}}$ may be a different member of $Y_{\tau,c}$. Cover exactingness is designed to tolerate precisely this motion inside an invariant small family. Replacing that family by the singleton $\{y\}$ would invalidate the argument.

The finite-change device is related to the name families in Blue–Goldberg’s construction.[2, proof of Theorem 3.5] Using just the explicitly defined finite-change class has a useful advantage here: we do not need a converse theorem classifying *all* Prikry generics that yield the same extension.

5 The Prikry consistency refinement

5.1 A localized shift, with fixed name parameters

For the rest of this section, $j : V_\Lambda \rightarrow V_\Lambda$ is an $\text{I3}_{\text{wf}(0)}$ embedding, $\kappa = \kappa_0$, and U_0 is the derived normal measure on κ . The following localization is the bridge from the iteration interface to the two lemmas just proved.

Lemma 5.1 (Coded localization and parameter fixedness). Let N_0 be a transitive set of cardinality κ satisfying a fixed, sufficiently strong finite fragment of ZFC, with

$$V_\kappa \cup \{\kappa, \bar{U}_0\} \subseteq N_0, \quad \bar{U}_0 = U_0 \cap N_0, \quad \text{Ord} \cap N_0 < \kappa^+.$$

Suppose N_0 regards $\bar{U}_0$ as a normal measure on κ . Then there are a transitive set N , an elementary map $i : N \rightarrow N$, and the corresponding map $J = j_{0,\omega}$ on N_0 -parameters such that

1. $N = J(N_0)$ is well-founded, $V_\Lambda \cup \{\Lambda\} \subseteq N$, and $i \upharpoonright \Lambda = j \upharpoonright \Lambda$;
2. $D = J(\bar{U}_0) \in N$ is a normal measure on Λ in N ;
3. $i(J(a)) = J(a)$ for every $a \in N_0$;
4. $c = \langle \kappa_n : n < \omega \rangle$ is Prikry-generic over N for D , and i lifts to an elementary map

$$e : W = N[G_c] \longrightarrow W \quad \text{with} \quad e(c) = c^-, \quad e \upharpoonright N = i.$$

In particular, W contains the full ambient V_Λ and regards Λ as an infinite cardinal with $|V_\Lambda| = \Lambda$.

Proof. We spell out the parameter issue in the well-founded coding construction used in the proof of [3, Theorem 5.4]. It matters because the Prikry name selected later need not be ordinal definable.

Choose a bijection $b_0 : \kappa \rightarrow N_0$ satisfying

$$b_0(0) = \kappa, \quad b_0(1) = \bar{U}_0, \quad b_0(\omega \cdot (\beta + 1)) = \beta \quad (\beta < \kappa). \quad (7)$$

The reserved ordinal slots are disjoint from $0, 1$; both their complement and the remaining values have cardinality κ . This permits completion to a bijection. The successor in $\beta + 1$ avoids a collision between the slot for the ordinal 0 and the slot for κ . Let

$$\xi E_0 \eta \iff b_0(\xi) \in b_0(\eta).$$

The sets N_0, b_0, E_0 have rank below κ^+ and hence belong to V_Λ . Define, in the direct limit,

$$E = J(E_0), \quad N = J(N_0), \quad b = J(b_0), \quad D = J(\bar{U}_0).$$

If $\delta = \text{Ord} \cap N_0$, the hypothesis (2) puts $J(\delta)$ in the well-founded part. The internal rank function of the transitive set N is bounded by this genuine ordinal. Thus N and its membership relation are well-founded, and their collapse is legitimate. We identify N with that collapse. The iteration interface gives $V_\Lambda \cup \{\Lambda\} \subseteq N$.

Now $b : (\Lambda, E) \cong (N, \in)$, and $j^+(E) = E$ by Proposition 2.3. Consequently

$$i = b \circ (j \upharpoonright \Lambda) \circ b^{-1} : N \rightarrow N$$

is elementary. Applying J to (7) and using the absoluteness of ordinal arithmetic gives

$$b(\omega \cdot (\beta + 1)) = \beta \quad (\beta < \Lambda).$$

Therefore $i(\beta) = j(\beta)$ for $\beta < \Lambda$.

Here is the additional parameter observation. Given any $a \in N_0$, choose $\xi < \kappa$ with $b_0(\xi) = a$. Since $J(\xi) = j(\xi) = \xi$,

$$J(a) = b(\xi), \quad i(J(a)) = b(j(\xi)) = b(\xi) = J(a). \quad (8)$$

Thus the shift fixes *every image parameter* from N_0 , not merely Λ and D . This does not make i the identity on N : $J''N_0$ need not be all of N .

Elementarity gives that D is a normal measure on Λ in N , with $D = N \cap J(U_0)$. By (3), every member of D contains a tail of c . The Mathias criterion makes G_c generic over N . The tail c^- is also generic and generates the same extension. Since $i(\mathbb{Q}) = \mathbb{Q}$ for $\mathbb{Q} = \mathbb{P}_D^N$ and $i''G_c \subseteq G_{c^-}$, the usual name interpretation defines the lift

$$e(\sigma^{G_c}) = i(\sigma)^{G_{c^-}}. \quad (9)$$

The forcing theorem proves that this is well-defined and elementary. The canonical generic-sequence name is fixed by i , so $e(c) = c^-$.

Prikry forcing preserves Λ as a cardinal in $N[G_c]$ and adds no bounded subsets of it. Also $V_\Lambda^{\text{ambient}} \subseteq N \subseteq W \subseteq V$, so the rank V_Λ is the same in W and the ambient universe. Each bounded rank has cardinality below Λ in W : choose a larger κ_n and use a bounding bijection of rank below Λ , which already belongs to W . Since $c \in W$ is cofinal in Λ , Choice in W gives $|V_\Lambda|^W = \Lambda$. $\square$

5.2 Homogeneity and one named counterexample

For later use define the first-order formula

$$\begin{aligned} \text{Bad}(\kappa, \alpha, y) \iff & \alpha > \kappa \wedge y \in V_\alpha \wedge \\ & \forall Y \subseteq V_\alpha [(y \in Y \wedge |Y| \leq \kappa) \implies \neg \text{RelEx}(\kappa, \alpha, Y)]. \end{aligned} \quad (10)$$

For a cardinal κ , $\neg \text{CEX}_\kappa(\kappa)$ is equivalent to $\exists \alpha \exists y \text{Bad}(\kappa, \alpha, y)$. Prikry forcing preserves κ as a cardinal, so this formulation applies in the extensions under consideration.

We need only the following familiar consequence of Prikry homogeneity: a sentence about the extension using ground-model parameters, but not a name for the chosen generic, is decided by the top condition. For completeness, below two Prikry conditions one can shrink the measure-one parts to a common set above both stems. Their cones are isomorphic by replacing the fixed stem and retaining the appended tail. The isomorphism preserves canonical names for ground-model parameters. Opposite decisions of such a sentence would therefore be impossible. This cone-homogeneity argument is enough; no particular global permutation of the entire forcing is needed.

If the top condition forces failure of cover exactingness, the least height α of failure is definable in the extension from κ . Some condition decides its ordinal value. Applying the preceding homogeneity fact to that value shows that a single ground-model ordinal α is forced to be the least bad height by the top condition. The maximum principle then supplies a name τ with

$$\mathbf{1}_{\mathbb{P}_{U_0}} \Vdash \text{Bad}(\check{\kappa}, \check{\alpha}, \tau). \quad (11)$$

The name τ itself need not be invariant under forcing isomorphisms. Its later invariance under the *localized shift* follows instead from (8).

5.3 The complete forcing argument

Theorem 5.2. If $j : V_\Lambda \rightarrow V_\Lambda$ is an $\text{I3}_{\text{wf}(0)}$ -embedding, $\kappa = \text{crit}(j)$, and U_0 is the derived normal measure, then

$$\mathbf{1}_{\mathbb{P}_{U_0}} \Vdash \text{CEX}_{\check{\kappa}}(\check{\kappa}). \quad (12)$$

Proof. Suppose otherwise. Homogeneity and the preceding discussion supply a height α and a name τ satisfying (11). Choose a reflecting rank V_ζ above Λ, α and the ranks of the names and forcing data.

Require reflection of (11), of the fixed finite fragment needed for the preceding lemmas, and of the forcing facts used in them. Inside the ambient universe choose

$$H < V_\zeta, \quad |H| = \kappa, \quad V_\kappa \cup \{\kappa, U_0, \alpha, \tau\} \subseteq H.$$

Let $\pi : H \cong N_0$ be the transitive collapse. It fixes V_κ and κ . Put

$$\bar{U}_0 = \pi(U_0), \quad a_0 = \pi(\alpha), \quad t_0 = \pi(\tau).$$

For subsets of κ that belong to H , the collapse is the identity. It follows that $\bar{U}_0 = N_0 \cap U_0$. The ordinal height of N_0 has cardinality at most κ and is below κ^+ . Thus Lemma 5.1 applies.

Let $J = j_{0,\omega}$ and use the resulting N, i, D, c, W, e . Set

$$a = J(a_0), \quad t = J(t_0), \quad \mathbb{Q} = \mathbb{P}_D^N.$$

The parameter fixedness clause gives

$$e(a) = a, \quad e(t) = t, \quad e(\mathbb{Q}) = \mathbb{Q}, \quad e(\Lambda) = \Lambda. \quad (13)$$

Elementarity of the collapse and of the direct-limit map transfers (11) to

$$N \models \mathbf{1}_\mathbb{Q} \Vdash \text{Bad}(\check{\Lambda}, \check{a}, t).$$

This is a statement about the *internal* forcing relation of N . Since G_c is N -generic, its forcing theorem gives

$$W \models \text{Bad}(\Lambda, a, t^{G_c}). \quad (14)$$

Apply Lemma 4.1 to t and c . In W it produces the set

$$Y = \{t^{G_h} : h \sim c\}, \quad t^{G_c} \in Y \subseteq V_a^W, \quad |Y|^W \leq \Lambda.$$

Here and below the sequences in the cloud are strictly increasing. Equations (13) and $e(c) = c^-$ imply $e(Y) = Y$. The hypotheses of Lemma 3.1 now hold for

$$(W, e, \lambda, c, \alpha, Y) = (W, e, \Lambda, c, a, Y).$$

In particular, $e(\kappa_n) = \kappa_{n+1}$, the full ambient V_Λ is in W , and W regards its size as Λ . Therefore

$$W \models \text{RelEx}(\Lambda, a, Y).$$

This contradicts (14), because the same Y belongs to W , contains t^{G_c} , and has size at most Λ there. The contradiction proves (12). $\square$

Where the strengthening occurs

The localized iteration alone provides an external self-embedding of W . The cloud lemma turns an arbitrary name value into a small invariant predicate, and the enumerated-domain tree turns preservation of that predicate into an internal witness at the required height. Neither “the embedding exists externally” nor “the name has a fixed code” would suffice by itself.

6 Consistency consequences and exact limits

6.1 A set-model conclusion

Corollary 6.1. The existence of an $I3_{\text{wf}(0)}$ -embedding implies the existence of a countable transitive model of

$$\text{ZFC} + \exists \lambda \text{ CEx}_\lambda(\lambda).$$

Consequently,

$$\text{Con}(\text{ZFC} + I3_{\text{wf}(0)}) \implies \text{Con}(\text{ZFC} + \exists \lambda \text{ CEx}_\lambda(\lambda)).$$

Proof. For a normalized rank-into-rank embedding, V_Λ is a model of ZFC. We use the rank-model passage from [3, proof of Corollary 5.5], checking the additional cover witnesses explicitly below. The forcing $\mathbb{P}_{U_0}$ and all its subsets have rank below Λ . A generic over V_Λ is therefore generic over V , and

$$V_\Lambda[G] = V[G]_\Lambda.$$

By Theorem 5.2, the latter rank model satisfies $\text{CEx}_\kappa(\kappa)$. To check the downward passage explicitly, a witness at height $\alpha < \Lambda$ consists of $Y \subseteq V_\alpha$, $X \subseteq V_\alpha$, and the graph of a map between subsets of V_α . Their ranks are at most α plus a fixed finite amount, which is below Λ . The same is true of a bijection certifying $|Y| \leq \kappa$. Set-structure satisfaction is absolute. Thus the witnesses can be retained in $V[G]_\Lambda$.

It follows that V_Λ satisfies the internal forcing assertion that its top condition forces $\text{CEx}_\kappa(\kappa)$. Choose a countable $A < V_\Lambda$ containing U_0, κ and collapse it to a transitive model M . This forcing assertion transfers to M . A generic filter for its collapsed forcing exists in the ambient universe by enumerating the countably many dense sets of M . Its extension is a countable transitive model of the displayed theory.

This is an ordinary relative-consistency argument formalizable over the stated theories. It does not require an external assumption that an arbitrary consistent theory already has a transitive model. $\square$

6.2 The revised strength interval

Use $A \geq_{\text{con}} B$ for the indicated relative-consistency reduction, and use strict comparison only where it is supplied by the established iteration hierarchy. The result yields

$$\begin{aligned} I2 &>_{\text{con}} I3_{\text{wf}(0)} \geq_{\text{con}} \exists \lambda \text{ CEx}_\lambda(\lambda), \\ \exists \lambda \text{ CEx}_\lambda(\lambda) &\geq_{\text{con}} \exists \lambda \text{ Ex}(\lambda) >_{\text{con}} I3. \end{aligned} \tag{15}$$

The strict comparisons and the ordinary-exacting lower bound are those of the existing hierarchy.[3, Theorem 5.10 and Corollary 5.11] The two middle comparisons are *not* claimed to be strict, and no equiconsistency between exactingness and cover exactingness follows.

In particular, the least-budget cover principle is placed strictly between $I3$ and $I2$ in this consistency-strength sense. The additional uniform cover requirement has not forced the upper bound back up to $I2$.

6.3 The cardinal-ordering issue survives unchanged

The argument does not preserve a specified strongly compact cardinal below κ through the Prikry construction. It therefore does not answer whether a cover-exacting cardinal can lie above a strongly compact cardinal. That is the explicit question in the 2026 notes.[2, Problem 5.2]

It also does not conflict with the cover-exacting obstruction above an extendible. If that obstruction is used, any model produced here has no extendible below the cover-exacting witness. The positive construction is about isolated existence, not the forbidden ordering of two cardinals. A statement that the forcing preserves “large cardinals” without specifying their position and preservation theorem would conceal exactly the missing step.

Finally, there is no claim of a new equiconsistency for ultraexactingness. The proof internalizes some branch of a tree; it does not show that the chosen local embedding has its restriction to V_λ in its domain. The ultraexacting clause cannot be read into the conclusion.

7 A definable-family inconsistency theorem

7.1 The only large-cardinal input

For λ exacting, the established regularity theorem says

$$\text{cf}^V(\lambda) = \omega, \quad \text{HOD}_{V_\lambda} \models “\lambda \text{ is a regular cardinal.}” \quad (16)$$

The second assertion is Theorem 2.10 of Aguilera–Bagaria–Lücke.[5] We do not reproduce its exacting-embedding proof. Everything in the present section after (16) is an explicit deduction from it.

Write $N = \text{HOD}_{V_\lambda}$. A set in OD_{V_λ} need not belong to N : its members might not be definable from permitted parameters. For example, a definable set of cofinal maps may have no individually definable member. On the other hand, an OD_{V_λ} set of ordinals, or an OD_{V_λ} graph between ordinals, does belong to N . Its transitive closure adds only hereditarily definable finite codes and ordinals. This observation is the bridge used below.

Theorem 7.1 (Projection width). Let λ be exacting, $0 < \rho < \lambda$, and let $F \in \text{OD}_{V_\lambda}$ be a nonempty set of cofinal maps $f : \rho \rightarrow \lambda$. Define

$$P_\xi = \{f(\xi) : f \in F\} \quad (\xi < \rho).$$

There is $\xi < \rho$ with P_ξ unbounded in λ . For any such ξ ,

$$\text{ot}(P_\xi) = \lambda, \quad |P_\xi|^V = \lambda, \quad |F|^V \geq \lambda. \quad (17)$$

If all members of F are increasing, the first two equalities hold for every coordinate greater than or equal to some $\xi < \rho$.

Proof. Fix one definition of F using ordinal parameters and a finite tuple $\vec{x}$ of members of V_λ . All definitions in this proof use that same tuple, together with ordinals. This uniformity is important. Each P_ξ is a set of ordinals definable from these parameters, and therefore belongs to N .

Suppose every P_ξ were bounded in λ . Define

$$b(\xi) = \sup\{\beta + 1 : \beta \in P_\xi\} \quad (\xi < \rho).$$

Each value is below λ . The *entire graph* of b is definable from $\vec{x}$ and the fixed ordinal parameters by substituting the definition of F . It is a graph between ordinals, so $b \in N$. Since $\rho < \lambda$ and λ is regular in N ,

$$\sup_{\xi < \rho} b(\xi) < \lambda.$$

Every $f \in F$ is pointwise bounded by b , contradicting the cofinality of f . Thus some P_ξ is unbounded. The increasing enumeration of this P_ξ is definable from P_ξ and belongs to N . If its order type were below λ , it would be a short cofinal map in N , again contradicting regularity. A subset of the ordinal λ has order type at most λ , so its order type is exactly λ . This is an absolute order-type computation. Since λ is also an ambient cardinal, $|P_\xi|^V = \lambda$. The evaluation map $F \rightarrow P_\xi$ is onto, so ambient Choice gives $|F|^V \geq \lambda$.

Finally, suppose all $f \in F$ are increasing and P_ξ is unbounded. For every $\eta \geq \xi$ and every $\beta < \lambda$, choose $f \in F$ with $f(\xi) > \beta$. Then $f(\eta) \geq f(\xi) > \beta$. Hence P_η is unbounded as well, and the same argument applies. $\square$

Corollary 7.2 (Small-family incompatibility). The following theory is inconsistent:

$$\text{ZFC} + \exists \lambda \exists \rho \exists F [\text{Ex}(\lambda) \wedge 0 < \rho < \lambda \wedge F \in \text{OD}_{V_\lambda} \wedge 0 < |F| < \lambda \\ \wedge \forall f \in F (f : \rho \rightarrow \lambda \text{ is cofinal})].$$

No definable selection of a member of F is required.

7.2 A local covering formulation

Definition 7.3. Let $\text{SmallODCover}(\lambda)$ assert that every $x \in V_{\lambda+1}$ belongs to a set $A \in \text{OD}_{V_\lambda}$ of ambient cardinality less than λ .

This is a definability covering principle, not the embedding-relative cover-exacting property. In particular, it must not be substituted for Definition 2.1 merely because both use the word “cover.”

Corollary 7.4.

$$\text{ZFC} \vdash \text{Ex}(\lambda) \implies \neg \text{SmallODCover}(\lambda).$$

More locally, if $c : \omega \rightarrow \lambda$ is cofinal and λ is exacting, every $A \in \text{OD}_{V_\lambda}$ with $c \in A$ has $|A| \geq \lambda$.

Proof. Given such an A , restrict it to

$$F = \{f \in A : f : \omega \rightarrow \lambda \text{ is cofinal}\}.$$

This family is nonempty, belongs to OD_{V_λ} , and is a subset of A . Theorem 7.1 implies $|A| \geq |F| \geq \lambda$. Since (16) supplies a cofinal c of rank λ , we have $c \in V_{\lambda+1}$ and the covering principle fails. $\square$

7.3 What this changes in the negative programme

The supplied report sought a single cofinal map in HOD or HOD_{V_λ} . [1, Section 5.4] The new deduction shows that a smaller demand suffices: construct a *definable family of candidates* of size below λ . No canonical least member and no definable well-order of the family are necessary. Taking coordinate suprema extracts the contradiction without making a choice inside the definability model.

This is nevertheless a conditional inconsistency theorem. We have not shown that strong compactness or extendibility below an ordinary exacting cardinal produces such a family. It would be incorrect to replace the missing family-construction theorem by a choice of candidates under an arbitrary ambient well-order.

The cloud construction does not violate this result. Its family may have size exactly λ , and its definition uses the generic sequence and ground-name parameters, not necessarily parameters in V_λ . Both distinctions are substantive. The positive argument requires a small invariant predicate; the negative argument requires a still smaller family definable from the specified low-rank parameters.

8 A complementary approximation obstruction

The regularity gap also yields a useful diagnostic for proposed covering or approximation arguments. This is an elementary fresh-sequence consequence, not a separate large-cardinal lower-bound theorem.

Proposition 8.1 (A fresh countable cofinal set). Let λ be exacting, let $N = \text{HOD}_{V_\lambda}$, and choose an increasing cofinal set $c \subseteq \lambda$ of order type ω in V . Then $c \notin N$, but

$$c \cap a \in N \tag{18}$$

for every $a \in N$ for which N contains an injection $a \rightarrow \rho$ for some $\rho < \lambda$. In fact, each intersection in (18) is finite.

Proof. If $c \in N$, its increasing enumeration would be a countable cofinal map in N , contrary to regularity of λ there. Now take a as in the statement. The set $a \cap \lambda$ belongs to N and is well-orderably smaller than λ there, hence is bounded in λ . An increasing cofinal sequence of order type ω has only finitely many points below any fixed bound less than λ . Thus $c \cap a$ is a finite set of ordinals and belongs to N . $\square$

To avoid concealing a choice assumption, call the following the *well-orderable δ -approximation test*: whenever $A \subseteq N$ is an ambient set and $A \cap a \in N$ for every $a \in N$ admitting an injection in N into some ordinal below δ , then $A \in N$. Proposition 8.1 refutes this test for every infinite $\delta \leq \lambda$, with the same witness c . When the inner model satisfies Choice, this is the usual small-set form of the approximation test. No blanket assertion of Choice for HOD_{V_λ} is used here.

This explains a limitation on one possible attack from the original report. An approximation principle strong enough to recover c cannot simply be assumed as a harmless background regularity property of the HOD pair: in the exacting configuration it is already a contradictory extra condition. A successful argument must derive that principle from some other stated hypothesis, not from the existence of the outer universe's cofinal sequence alone.

9 Proof audit and the remaining research targets

9.1 The interfaces checked in the construction

Potential failure	How it is addressed
An ill-founded direct limit	Only the coded transitive set $J(N_0)$ is used. Its height is $J(\delta)$ for $\delta < \kappa^+$, exactly where the $\text{I3}_{\text{wf}(0)}$ hypothesis guarantees well-foundedness.
An arbitrary name is not fixed	Equation (8) fixes every $J(a)$ with $a \in N_0$. It does not require a to be ordinal definable.
The generic sequence is not fixed	It moves to its tail. The family is invariant under finite changes of its range, so the family, rather than its chosen member, is fixed.
The predicate has the wrong rank	The name is forced to belong to V_a over N . All finite-change generics generate the same W , so every evaluation lies in the same V_a^W .
The cardinal bound varies with height	The family of finite modifications has size at most Λ in W , independently of the name and of a . This is the transported version of the single bound κ .

Potential failure	How it is addressed
An external embedding is used as an internal witness	The functions q_n have rank below Λ and belong to W . They code high-rank partial maps; a tree in W then has an internal branch by ordinal-rank absoluteness.
An image set is confused with a pointwise image	$e(Y) = Y$ follows by elementarity of the defining formula. The proof does not assert $e''Y = Y$.
A global scheme is assumed in a small rank	The collapse is taken from a rank reflecting a fixed finite collection of formulas. Appendix A identifies their roles.
A set of definable candidates is treated as hereditarily definable	The width proof only puts the ordinal projections and their uniformly defined bound function into HOD_{V_λ} . It never puts all of F there.
A consistency bound becomes equiconsistency	Only the displayed one-way reductions are claimed. No reverse arrow from cover exactingness to $\text{I3}_{\text{wf}(0)}$ is established.

9.2 Three focused next questions

The first question concerns the middle of the revised interval:

Question 9.1. Can the $\text{I3}_{\text{wf}(0)}$ upper bound be lowered further for $\exists \lambda \text{ CEx}_\lambda(\lambda)$, or can the cover property supply an iteration lower bound not already implied by ordinary exactingness?

The construction identifies a specific possible obstruction to lowering it: the chosen counterexample hull must remain well-founded after iteration. One would need either a smaller well-foundedness certificate for those particular hulls or a different internalization setup.

Question 9.2. Under a smaller strongly compact cardinal, can one construct an OD_{V_λ} family of fewer than λ candidates containing one short cofinal map into λ ?

An affirmative derivation from the exacting–strongly-compact hypotheses would yield an inconsistency through Corollary 7.2. The family bound must be *strictly* below λ , and the allowed parameters must be verified. This question is a sufficient local route, not an assertion that the missing construction follows from compactness.

Question 9.3. Can a cover-exacting construction retain a specified strongly compact cardinal below the cover-exacting witness, while avoiding the club-definability obstruction proved for extendibles?

The present construction gives no such preservation theorem. Its successful isolated-existence case is a control against arguments that would refute cover exactingness without using the smaller cardinal.

The global cardinal-preserving and Icarus tracks are not resolved by these results. In particular, nothing here turns internal stationarity in a target inner model into ambient stationarity. The original report’s stationary-collision diagnostic still requires that separate input.

9.3 Overall conclusion

The continuation produces a positive consistency refinement and a negative local criterion that work in opposite directions without conflict. A Prikry name can be placed in a family of at most the critical limit’s cardinality whose defining class is invariant under the shift. A low-rank-parameter-definable family of *fewer* than that many cofinal maps is impossible at an exacting cardinal.

A Finite fragments, ranks, and the model ledger

A.1 Why the small models have enough set theory

A transitive collapse of a hull in V_ζ must not be called a model of all of ZFC merely because ζ was chosen large. The proof uses the following standard finite-fragment procedure instead.

First fix the formulas defining the forcing relation for $\text{Bad}(\kappa, \alpha, y)$ and its negation, the set-satisfaction relation, Prikry-name evaluation, the finite-change family, and the tree of finite partial maps. Include the finite collection of ZFC consequences used to construct these objects and to carry out their proofs. In particular, include the needed instances of well-founded recursion and of the forcing theorem, normal-measure Prikry genericity, the elementary hull construction, and the assertion that a well-founded set relation has an ordinal rank.

Each of these is a uniform assertion or uses finitely many instances in a fixed formal proof. There is therefore a finite subtheory of ZFC sufficient for the argument. Enlarge the reflecting list to contain this theory and the displayed forced-counterexample statement. Lévy reflection supplies a rank V_ζ satisfying and correctly reflecting this finite list, with ζ above all specified parameters. Its elementary hull and transitive collapse satisfy the same list. The direct-limit map transfers it to the well-founded N , and its included forcing theorems apply to $N[G_c]$.

This is a finite-proof reflection argument, not a demand for $V_\zeta < V$ for all formulas simultaneously. The internalization lemma quantifies over all first-order formulas of *set structures* using their single set-satisfaction relation; it does not require a truth predicate for the universe.

A.2 The universe changes that must not be conflated

Model	Role
V	Original ambient universe, containing j and the whole construction of the local auxiliary model.
$V[G]$	Hypothetical Prikry extension in which failure at κ is assumed, in order to obtain the forced counterexample.
N_0	Small transitive collapse of a hull of a reflecting V_ζ ; its cardinality is κ and its ordinal height is below κ^+ .
M_ω^j	Possibly ill-founded direct limit. It is used as a transport structure, not as the final universe.
$N = J(N_0)$	Well-founded, transitive, sufficiently correct small model containing the ambient V_Λ .
$W = N[G_c]$	Auxiliary model inside the original V . Its external shift is internalized only through a tree argument. It is not $V[G]$.
$M[H]$	Countable transitive model of the final theory obtained at the end of Corollary 6.1.

The counterexample is transported by

$$V \models \mathbf{1} \Vdash \text{Bad}(\kappa, \alpha, \tau) \implies N \models \mathbf{1} \Vdash \text{Bad}(\Lambda, a, t) \implies W \models \text{Bad}(\Lambda, a, t^{G_c}).$$

The contradiction is entirely *inside* W . It then rules out the initial forced failure and proves the result about $V[G]$.

A.3 Rank checks

The membership relation E_0 on κ is a subset of $\kappa \times \kappa$ and belongs to $V_{\kappa+1}$, using the usual finite-rank coding of ordered pairs. The coding bijection and the transitive collapse have rank below $\kappa^+ < \Lambda$. The image relation E is a subset of $\Lambda \times \Lambda$, with rank at most Λ , so it belongs to $V_{\Lambda+1}$.

For internalization, $q_n = e \restriction \kappa_n$ is a subset of $\kappa_n \times \kappa_{n+1}$. Its rank is below Λ , so $q_n \in W$ follows from rank inclusion, not from a closure principle for W at length Λ . The actual partial map p_n may have much higher rank, since its domain lies in V_a^W . It is nevertheless in W because it is constructed there from q_n, f, g .

A cofinal map $c : \omega \rightarrow \lambda$ has rank λ and belongs to $V_{\lambda+1}$, not V_λ . This is why its use as a parameter in the cloud definition is not permitted in the negative theorem's OD_{V_λ} family hypothesis.

B Source and verification record

The supplied `Large_Cardinals_Unified_Report.tex` was read as the starting research document. Its palette, font packages, page geometry, heading styles, theorem styling, and assessment-box design are retained here. The new text is a continuation rather than an alteration of the original report's claims.

The primary sources checked for the main argument are the July 2026 cover-exacting notes, version 1 of *Large cardinals beyond HOD*, version 2 of the iterability paper, and version 4 of the original exacting-cardinal paper. The cover notes explicitly acknowledge that related consistency constructions yield cover variants; this is why historical novelty is not inferred from the absence of the exact $\text{I3}_{\text{wf}(0)}$ statement in the supplied report.

The proof audit concentrated on the fixedness of arbitrary transported name parameters, the finite-change equivalence relation, the internal cardinality bound, the well-founded part of the iterate, and the internal existence of the final embedding. No finite computation is offered as evidence for the infinite set-theoretic assertions. The manuscript has not been formalized in Lean or another proof assistant, and it has not undergone independent specialist review.

The archive includes the compilable source and rendered PDF, with a build script and a short dependency ledger. No external figure assets or font files are needed in the archive. The original Palatino-style text and mathematical typography are selected by the same `newpxtext` and `newpxmath` packages as the supplied source.

References

- [1] *Large cardinals at the inconsistency frontier: a unified assessment of ZFC-compatible hypotheses, their obstructions, and the evidence on both sides*. Supplied research report prepared for Vladimir, 18 September 2026. Source file: `Large_Cardinals_Unified_Report.tex`. The starting document, particularly its sections on exactingness, cover exactingness, and the proposed research programme.
- [2] G. Goldberg. *Consistency beyond the Kunen inconsistency*. Lecture notes dated 1 July 2026, reporting joint work with D. Blue. <https://math.berkeley.edu/~goldberg/Slides/CoverExact.pdf>. Definitions 2.6 and 3.1; Theorem 3.5 and the subsequent remark; Theorem 3.6; Problem 5.2. Used as lecture notes, not described as a refereed publication.
- [3] J. P. Aguilera, J. Bagaria, G. Goldberg, and P. Lücke. *Large cardinals beyond HOD*. [arXiv:2509.10254v1](https://arxiv.org/abs/2509.10254), 12 September 2025. Proposition 5.1, Definition 5.2, the proof of Theorem 5.4, Corollary 5.5, Theorem 5.10, and Corollary 5.11 are the principal locators for the iteration-localization and consistency comparisons.
- [4] A. Andretta and V. Dimonte. *The iterability hierarchy above I_3* . *Archive for Mathematical Logic* **58** (2019), 77–97. [doi:10.1007/s00153-018-0624-5](https://doi.org/10.1007/s00153-018-0624-5); [arXiv:1712.03877v2](https://arxiv.org/abs/1712.03877). Section 3, especially Lemmas 3.3–3.5, supplies the iteration and amenable-predicate elementarity background.
- [5] J. P. Aguilera, J. Bagaria, and P. Lücke. *Large cardinals, structural reflection, and the HOD Conjecture*. [arXiv:2411.11568v4](https://arxiv.org/abs/2411.11568), 16 September 2025. Theorem 2.10 supplies regularity of an exacting cardinal in HOD_{V_λ} ; the conventions immediately preceding it specify OD_Γ and HOD_Γ .